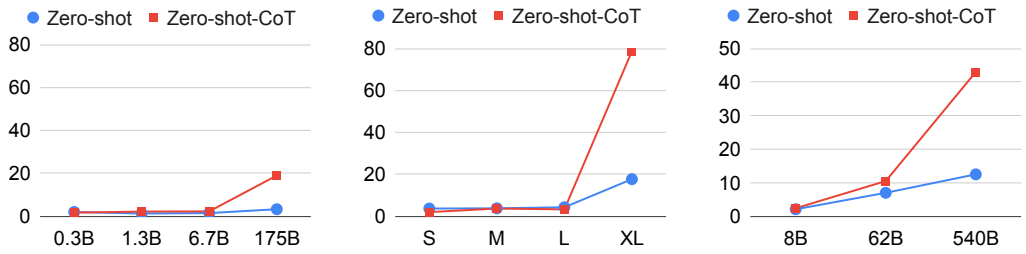

(a) MultiArith on Original GPT-3 (b) MultiArith on Instruct GPT-3 (c) GMS8K on PaLM

Figure 3: Model scale study with various types of models. S: text-ada-001, M: text-babbage-001, L: text-curie-001, XL: text-davinci-002. See Appendix A.3 and E for the detail.

Table 3: Examples generated by Zero-Shot-CoT on CommonsenseQA for Error Analysis.

Example 1 **(Reasoning is correct.)**	Example 2 **(Model cannot choose only one.)**
Q: Where is a well used toy car likely to be found? Answer Choices: (A) child's room (B) boy's bedroom (C) own home (D) toy store (E) house **A: Let's think step by step.** A toy car is likely to be found in a child's room. **A child's room is likely to be found in a house.** Therefore, a toy car is likely to be found in a house. **Therefore, among A through E, the answer is** E. ✗ **(GT : A, Zero-shot: A)**	**Q:** What would be necessary for getting in shape? Answer Choices: (A) good health (B) exercise (C) muscle tone (D) sweat (E) feel better **A: Let's think step by step.** In order to get in shape, you need to be able to do three things: 1. Exercise regularly 2. Eat a healthy diet 3. Get enough sleep If you can do all three of those things, then you will be well on your way to getting in shape! **Therefore, among A through E, the answer is B, C, and D.** **(GT : B, Zero-shot: B)**

Comparison with other baselines Table 2 compares the performances on two arithmetic reasoning benchmarks (MultiArith and GSM8K) across Zero-shot-CoT and baselines. The large gap between standard prompting (1st block) and chain of thought prompting (2nd block) suggests that these tasks are difficult without eliciting multi-step reasoning. Major improvements are confirmed on both Instruct GPT-3 (text-davinci-002) and PaLM (540B) models (4th block). While Zero-shot-CoT naturally underperforms Few-shot-CoT, it substantially outperforms standard Few-shot prompting with even 8 examples per task. For GSM8K, Zero-shot-CoT with Instruct GPT-3 (text-davinci-002) also outperforms finetuned GPT-3 and standard few-shot prompting with large models (PaLM, 540B), reported in Wei et al. [2022] (3rd and 4th block). See App. D for more experiment results with PaLM.

Does model size matter for zero-shot reasoning? Figure 3 compares performance of various language models on MultiArith / GSM8K. Without chain of thought reasoning, the performance does not increase or increases slowly as the model scale is increased, i.e., the curve is mostly flat. In contrast, the performance drastically increases with chain of thought reasoning, as the model size gets bigger, for Original/Instruct GPT-3 and PaLM. When the model size is smaller, chain of thought reasoning is not effective. This result aligns with the few-shot experiment results in Wei et al. [2022]. Appendix E shows extensive experiment results using wider variety of language models, including GPT-2, GPT-Neo, GPT-J, T0, and OPT. We also manually investigated the quality of generated chain of thought, and large-scale models clearly demonstrate better reasoning (See Appendix B for the sampled outputs for each model).

Error Analysis To better understand the behavior of Zero-shot-CoT, we manually investigated randomly selected examples generated by Instruct-GPT3 with Zero-shot-CoT prompting. See Appendix C for examples, where some of the observations include: (1) In commonsense reasoning (CommonsenseQA), Zero-shot-CoT often produces flexible and reasonable chain of thought even when the final prediction is not correct. Zero-shot-CoT often output multiple answer choices when the model find it is difficult to narrow it down to one (see Table 3 for examples). (2) In arithmetic

Table 4: Robustness study against template measured on the MultiArith dataset with text-davinci-002. (*1) This template is used in Ahn et al. [2022] where a language model is prompted to generate step-by-step actions given a high-level instruction for controlling robotic actions. (*2) This template is used in Reynolds and McDonell [2021] but is not quantitatively evaluated.

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	**78.7**
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		Abrakadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

Table 5: Robustness study of Few-shot-CoT against examples. When the examples are from entirely different tasks, the performance generally becomes worse, but when the answer formats are matched (i.e. CommonsenseQA to AQUA-RAT, multiple-choice), the performance loss is less severe. †CommonsenseQA samples are used in this variation

	Zero-shot	Few-shot-CoT †	Zero-shot-CoT	Few-shot-CoT
AQUA-RAT	22.4	31.9	33.5	39.0
MultiArith	17.7	27.0	78.7	88.2

reasoning (MultiArith), Zero-shot-CoT and Few-shot-CoT show substantial differences regarding the error patterns. First, Zero-shot-CoT tends to output unnecessary steps of reasoning after getting the correct prediction, which results in changing the prediction to incorrect one. Zero-shot-CoT also sometimes does not start reasoning, just rephrasing the input question. In contrast, Few-shot-CoT tend to fail when generated chain of thought include ternary operation, e.g. $(3 + 2) * 4$.

How does prompt selection affect Zero-shot-CoT? We validate the robustness of Zero-shot-CoT against input prompts. Table 4 summarizes performance using 16 different templates with three categories. Specifically, following Webson and Pavlick [2022], the categories include instructive (encourage reasoning), misleading (discourage reasoning or encouraging reasoning but in a wrong way), and irrelevant (nothing to do with reasoning). The results indicate that the performance is improved if the text is written in a way that encourages chain of thought reasoning, i.e., the templates are within "instructive" category. However, the difference in accuracy is significant depending on the sentence. In this experiment, "Let's think step by step." achieves the best results. Interestingly, it is found that different templates encourage the model to express reasoning quite differently (see Appendix B for sample outputs by each template). In contrast, when we use misleading or irrelevant templates, the performance does not improve. It remains an open question how to automatically create better templates for Zero-shot-CoT.

How does prompt selection affect Few-shot-CoT? Table 5 shows the performance of Few-shot-CoT when using examples from different datasets: CommonsenseQA to AQUA-RAT and CommonsenseQA to MultiArith. The domains are different in both cases, but the answer format

is the same in the former. Surprisingly, the chain of thought examples from different domains (common sense to arithmetic) but with the same answer (multiple-choice) format provide substantial performance gain over Zero-shot (to AQUA-RAT), measured relative to the possible improvements from Zero-shot-CoT or Few-shot-CoT. In contrast, the performance gain becomes much less when using examples with different answer types (to MultiArith), confirming prior work [Min et al., 2022] that suggests LLMs mostly leverage the few-shot examples to infer the repeated format rather than the task itself in-context. Nevertheless, for both cases the results are worse than Zero-shot-CoT, affirming the importance of task-specific sample engineering in Few-shot-CoT.

5 Discussion and Related Work

Table 6: Summary of related work on arithmetic/commonsense reasoning tasks. Category denotes the training strategy. CoT denotes whether to output chain of thought. Task column lists the tasks that are performed in corresponding papers. AR: Arithmetic Reasoning, CR: Commonsense Reasoning.

Method	Category	CoT	Task	Model
Rajani et al. [2019]	Fine-Tuning	✓	CR	GPT
Cobbe et al. [2021]	Fine-Tuning	✓	AR	GPT-3
Zelikman et al. [2022]	Fine-Tuning	✓	AR,CR	GPT-3, etc
Nye et al. [2022]	Fine-Tuning[5]	✓	AR	Transformer(Decoder)
Brown et al. [2020]	Few/Zero-Shot		CR	GPT-3
Smith et al. [2022]	Few/Zero-Shot		AR,CR	MT-NLG
Rae et al. [2021]	Few-Shot		AR,CR	Gopher
Wei et al. [2022]	Few-Shot	✓	AR,CR	PaLM, LaMBDA, GPT-3
Wang et al. [2022]	Few-Shot	✓	AR,CR	PaLM, etc
Chowdhery et al. [2022]	Few-Shot	✓	AR,CR	PaLM
Shwartz et al. [2020]	Zero-Shot	✓	CR	GPT-2, etc
Reynolds and McDonell [2021]	Zero-Shot	✓	AR	GPT-3
Zero-shot-CoT (Ours)	Zero-Shot	✓	AR,CR	PaLM, Instruct-GPT3, GPT-3, etc

Reasoning Ability of LLMs Several studies have shown that pre-trained models usually are not good at reasoning [Brown et al., 2020, Smith et al., 2022, Rae et al., 2021], but its ability can be substantially increased by making them produce step-by-step reasoning, either by fine-tuning [Rajani et al., 2019, Cobbe et al., 2021, Zelikman et al., 2022, Nye et al., 2022] or few-shot prompting [Wei et al., 2022, Wang et al., 2022, Chowdhery et al., 2022] (See Table 6 for summary of related work). Unlike most prior work, we focus on zero-shot prompting and show that a single fixed trigger prompt substantially increases the zero-shot reasoning ability of LLMs across a variety of tasks requiring complex multi-hop thinking (Table 1), especially when the model is scaled up (Figure 3). It also generates reasonable and understandable chain of thought across diverse tasks (Appendix B), even when the final prediction is wrong (Appendix C). Similar to our work, Reynolds and McDonell [2021] demonstrate a prompt, "Let's solve this problem by splitting it into steps", would facilitate the multi-step reasoning in a simple arithmetic problem. However, they treated it as a task-specific example and did not evaluate quantitatively on diverse reasoning tasks against baselines. Shwartz et al. [2020] propose to decompose a commonsense question into a series of information seeking question, such as "what is the definition of `[X]`". It does not require demonstrations but requires substantial manual prompt engineering per each reasoning task. Our results strongly suggest that LLMs are decent zero-shot reasoners, while prior work [Wei et al., 2022] often emphasize only few-shot learning and task-specific in-context learning, e.g. no zero-shot baselines were reported. Our method does not require time-consuming fine-tuning or expensive sample engineering, and can be combined with any pre-trained LLM, serving as the strongest zero-shot baseline for all reasoning tasks.

Zero-shot Abilities of LLMs Radford et al. [2019] show that LLMs have excellent zero-shot abilities in many *system-1* tasks, including reading comprehension, translation, and summarization.

[5]Nye et al. [2022] also evaluates few-shot settings, but the few-shot performances on their domains are worse than the fine-tuning results.